

WATSON LAKE, YT, Canada

WMO: 711990

Lat: 60.1142N

Lon: 128.8237W

Elev: 2242

StdP: 13.54

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.4	-39.9	-49.4	0.3	-43.9	-44.8	0.4	-36.2	16.4	13.3	13.6	13.4	0.6	280	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	79.1	58.8	75.3	57.0	71.8	55.6	60.5	74.8	58.8	71.4	57.3	68.7	6.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.6	71.4	62.8	53.8	66.8	61.5	52.1	62.7	60.5	27.7	75.4	26.6	71.5	25.5	68.8	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.5	13.1	DB	-52.1	85.1	6.1	3.2	-56.5	87.5	-60.1	89.4	-63.5	91.2	-67.9	93.5	(4)
(5)			WB	-48.1	64.7	7.7	3.1	-53.6	66.9	-58.1	68.8	-62.4	70.5	-68.0	72.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	-6.7	0.7	13.1	31.8	46.5	56.3	59.5	56.0	45.9	31.9	6.9	-4.6
	DBStd	26.52	18.01	14.28	14.45	8.74	7.51	5.89	4.63	5.79	5.95	8.34	13.77	15.47
	HDD50	8701	1757	1381	1143	547	160	8	1	11	147	563	1292	1691
	HDD65	13438	2222	1801	1608	997	576	271	181	283	573	1028	1742	2156
	CDD50	763	0	0	0	0	50	197	294	199	23	0	0	0
	CDD65	26	0	0	0	0	1	10	10	5	0	0	0	0
	CDH74	460	0	0	0	0	20	139	176	124	1	0	0	0
	CDH80	81	0	0	0	0	2	31	28	20	0	0	0	0
	WSAvg	4.1	1.8	2.3	4.3	5.3	5.6	5.5	5.1	4.9	4.8	4.7	2.7	2.0
	PrecAvg	17.2	1.2	1.0	0.7	0.6	1.5	2.1	2.4	1.9	1.9	1.5	1.1	1.3
(15) Precipitation	PrecMax	23.2	2.1	2.8	1.8	1.8	4.6	4.3	5.5	3.7	3.9	3.1	2.3	2.6
	PrecMin	12.2	0.5	0.0	0.0	0.1	0.2	0.3	0.3	0.4	0.6	0.6	0.4	0.2
	PrecStd	2.8	0.4	0.6	0.5	0.5	0.9	1.0	1.2	0.8	0.7	0.6	0.6	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	38.3	46.4	58.8	76.4	84.3	83.6	83.0	70.0	55.1	41.8	36.6
		MCWB	33.7	33.6	37.5	44.0	54.3	59.6	60.5	61.3	56.0	46.5	36.6	34.1
	2%	DB	30.3	32.2	41.9	52.8	71.1	77.8	78.8	77.5	63.0	49.8	35.9	31.0
		MCWB	28.5	29.6	34.3	40.5	51.4	57.6	59.0	59.0	51.0	42.4	32.3	28.8
	5%	DB	22.4	26.5	38.4	49.1	66.2	73.2	74.9	73.0	59.7	46.2	30.9	25.2
		MCWB	21.2	24.7	32.4	38.4	49.7	55.2	57.6	57.4	49.5	40.4	28.7	23.9
	10%	DB	14.6	21.6	34.9	45.7	61.4	69.2	71.4	68.9	56.9	43.1	25.7	16.2
		MCWB	13.7	20.3	30.3	36.4	47.2	53.6	56.3	55.8	48.3	38.3	24.4	15.4
	0.4%	WB	34.2	33.9	38.1	44.7	56.2	62.6	62.9	62.7	56.6	47.5	36.7	34.4
		MCDB	36.6	37.4	45.2	56.9	69.5	77.1	77.6	78.0	68.4	53.4	40.7	36.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.6	29.5	34.8	41.4	52.7	59.2	60.4	60.2	53.1	43.8	32.4	28.9
		MCDB	30.7	32.0	40.7	51.3	67.7	73.6	74.2	74.3	60.1	48.8	35.4	30.9
	5%	WB	21.8	24.7	32.7	39.0	50.4	56.9	58.9	58.4	51.0	40.9	28.7	23.9
		MCDB	23.0	26.4	38.0	47.8	64.6	70.2	71.0	70.4	57.3	45.3	30.7	25.3
	10%	WB	14.2	20.2	30.4	37.0	48.3	55.0	57.5	56.6	49.1	38.5	24.4	15.3
		MCDB	15.0	21.6	34.7	44.4	60.3	66.9	68.6	66.8	55.3	42.4	25.9	15.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	21.9	26.4	22.4	21.5	19.7	19.4	19.7	17.4	12.7	14.5	15.3
		MCDBR	21.2	23.1	25.8	26.1	29.4	26.8	27.2	28.2	22.9	17.5	17.6	21.7
		MCWBR	19.9	20.9	19.3	15.6	14.1	11.5	11.8	13.1	13.0	11.4	14.9	20.2
		MCDBR	21.3	22.0	24.6	23.5	26.5	23.6	23.6	24.4	19.6	16.3	16.7	21.7
	5% WB	MCWBR	19.9	20.1	18.8	14.6	13.3	10.8	11.0	12.2	11.9	11.2	14.2	20.0
(40) Clear-Sky Solar Irradiance	taub		0.213	0.240	0.263	0.326	0.353	0.369	0.369	0.355	0.312	0.273	0.219	0.186
	taud		2.164	2.319	2.289	2.173	2.363	2.380	2.401	2.460	2.566	2.551	2.196	2.021
	Ebn at Noon		190	253	280	275	277	273	270	266	263	241	184	147
	Edh at Noon		12	19	28	38	34	34	33	29	22	16	12	9
(44) All-Sky Solar Radiation	RadAvg		122	387	896	1432	1676	1730	1598	1284	806	388	145	68
	RadStd		12	42	72	93	132	103	116	79	74	34	18	6

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.92	+3.74	N/A	N/A	N/A	N/A	N/A	-373	+77	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page